

Fleet Intelligence for Autonomous HEMM Operations

Project Report

Autonomous Navigation Simulation Project

July 2026

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Chapter 1

Introduction and Problem Statement

Mines move material with fleets of haul trucks between **load zones** (where material is picked up) and **dump zones** (where it is emptied). The software choice of *which truck goes to which mine next* is **dispatch**. Bad dispatch wastes fuel and time; good dispatch raises completed hauls per hour and spreads work fairly across mines.

Problem. We already have a multi-truck simulator. What we lacked is a **data path** from those runs to evidence-based decisions: clean trip tables, forecasts of how long a haul will take (**ETA**), and a fair way to score alternate dispatch rules.

Core question. From logged simulation traces, can we measure operations, predict outcomes, and rank dispatch policies—without guessing?

1.1 What this project does

Table 1.1: Three analytics layers

Layer	Plain meaning	Technical name
Describe	Turn raw logs into trip and queue tables	KPI extraction
Predict	Estimate haul duration from context	ETA models
Prescribe	Score alternate rules on the same history	Offline policy replay

Scope. Simulation + offline analytics on a cleaned **~12-truck** trip set. Not a real-mine deployment, and not live A/B of new policies inside the closed-loop sim (future work).

Roadmap. Ch. 2 system → Ch. 3 data → Ch. 4 policy evaluation → Ch. 5 EDA/ETA preview → Ch. 6 conclusions and future work.

1.2 Motivating patterns in the trip data

Time and fuel are the two quantities every good dispatcher must trade off. The figures below show that trips are not one fixed duration, and that fuel tends to move with cycle time.

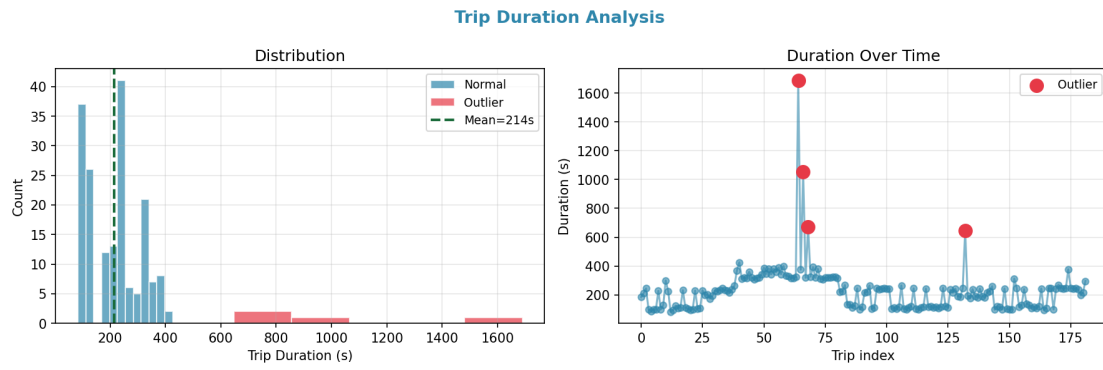


Figure 1.1: Trip duration structure and spread in logged hauls.

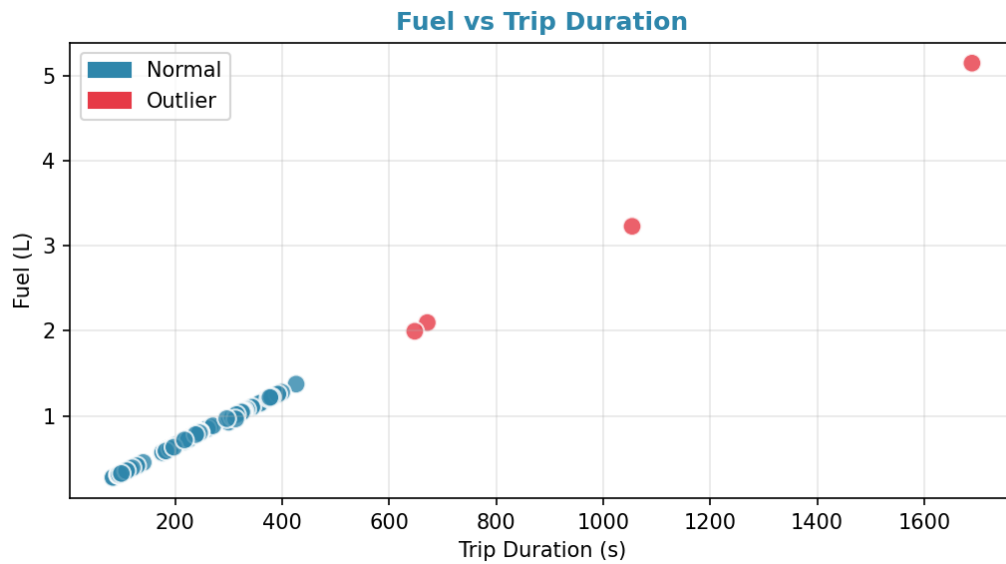


Figure 1.2: Fuel versus trip duration—motivation for joint productivity–efficiency metrics.

Chapter 2

System and Simulator

Think of the simulator as a **digital haul road**: trucks follow a mapped network, stop to load/unload, and get a new destination when free.

- **Map.** A **graph** (nodes = places, edges = road links) plus smooth **waypoint polylines** for motion. Cached in `map_cache.pkl` / `waypoints.pkl`.
- **Fleet.** Configured in `mine_config.json` (recent collection: **12 trucks**, capacity limits per mine).
- **A trip.** Empty truck → mine → load → dump → unload. One finished trip = one `UNLOAD_COMPLETE` event.
- **Dispatch.** A **dispatcher** proposes mines (global plan + local fallback). A **local planner** (BFS/A*) turns “go to mine X” into a node path, then a geometric path.
- **Motion.** Controllers track that path (MPC in GUI; lighter control when farming data headless). Soft tracking and lane offsets mean trucks may leave the painted road—so **continuous** metrics (fuel, path length) are noisier than **discrete** trip counts.
- **Logging.** Paired CSVs: **telemetry** (≈ 1 Hz) and **events** (arrivals, loads, unloads).

```
Sim -> logs (events + telemetry) -> merged tables
      -> trips / queues / KPIs -> models & policies
```

2.1 Road geometry drives cycle cost

The planner uses road-graph distances, not only straight-line shortcuts. Detour factors show how much longer the real road is than a Euclidean line.

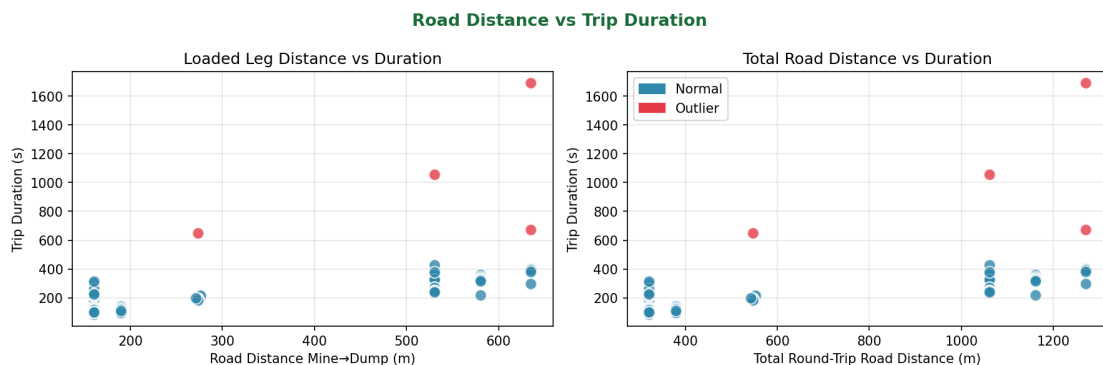


Figure 2.1: Road distance versus trip duration (geometry of the haul network).

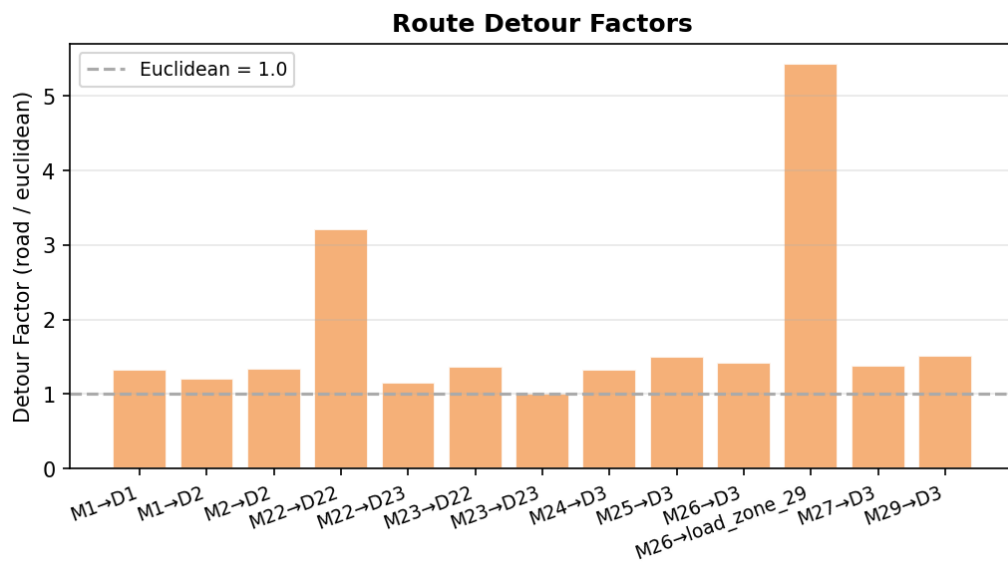


Figure 2.2: Detour factor: road distance relative to Euclidean distance.

Chapter 3

Data Collection and Cleaning

We do not analyze animations. We analyze **tables built from logs**.

3.1 What the data is

Table 3.1: Primary datasets

Dataset	One row is	Why it matters
events_*.csv	One operational fact	Defines trip boundaries
telemetry_*.csv	Periodic truck snapshot	Speed, fuel, position
merged_*.csv	All sessions aligned	Cross-run analysis
trips.csv	One completed haul cycle	Primary analysis unit
queue_events.csv	Congestion / idle bins	ETA & dispatch context
etc.		
trips_graph.csv	Trips + road distances	Geometry-aware features
trips_all_mixed.csv	Unfiltered archive	Audit trail

Definition: a **trip** = one UNLOAD_COMPLETE (full mine→dump cycle).

3.2 How the data was produced

1. Headless sims write session logs under `logs/`.
2. `merge_logs.py` concatenates sessions with time offsets.
3. `kpi_extractor.py -merged` builds `trips.csv` and KPI files.
4. Cleaning **filters** trips; it does not invent them.

Raw yield: **485** completed trips across **17** sessions (truck IDs 1–12 in the union; not every session used the full fleet).

3.3 Why size $\neq$ quality

Mixing 3-truck and 12-truck trips would make duration, queues, and dispatch scores **incomparable**. Final claims use **279** rows only.

3.4 Analysis dataset

Coverage: **10** load zones, **14** dump zones.

Table 3.2: Data slices

Slice	Trips	Trucks/session	Role
Excluded (early)	206	2–3	Light traffic; weak congestion
Kept for analysis	279	9–12	Full-fleet; policy-relevant
All mixed (archive)	485	mixed	Source before filter

Table 3.3: Sessions in `trips.csv` (279 trips)

Session	Trips	Trucks	Duration (s)	Fuel (L)
20260713_030256	71	11	350.4	1.2
20260713_161650	81	12	305.9	1.0
20260713_200004	87	12	362.1	1.2
20260714_011337	40	9	249.5	0.8
Total	279	12	326.7	1.076

3.5 What the cleaned trips look like

Same nominal route can still show large duration spread—a congestion and timing signal that only appears once enough full-fleet trips exist.

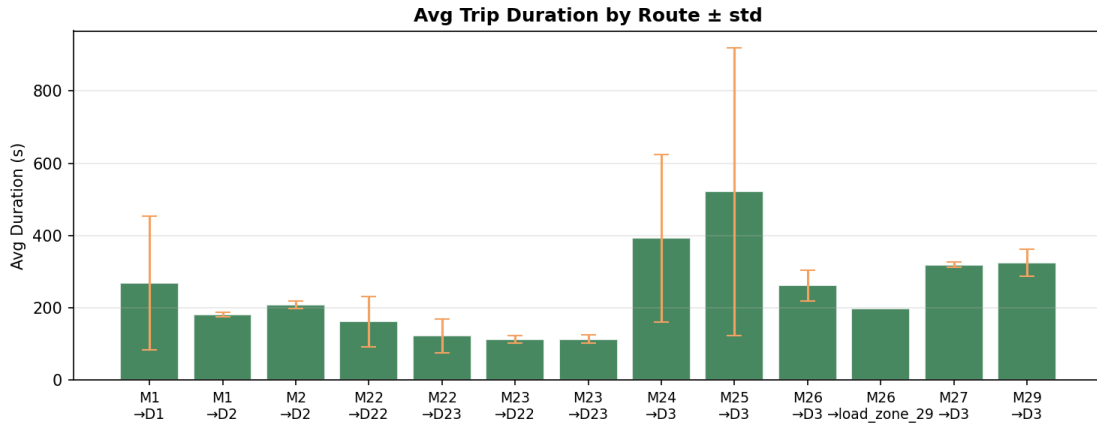


Figure 3.1: Duration by route: same mine–dump pair, different cycle times.

3.6 Chapter conclusions

1. Unit of analysis = completed trip (UNLOAD_COMPLETE).
2. Raw pool = 485; decision set = **279** full-fleet-era trips.
3. Motive: hold fleet scale constant so congestion and dispatch effects are real.
4. Policy and ETA chapters use this 279-row table (and `trips_graph.csv`).
5. **279 cleaner rows beat 485 mixed rows** for busy-fleet conclusions.

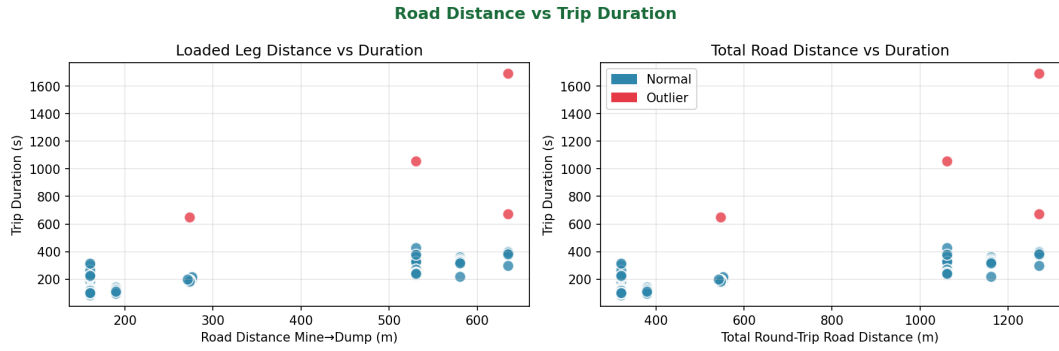


Figure 3.2: Longer road distance tends to mean longer trips in the analysis set.

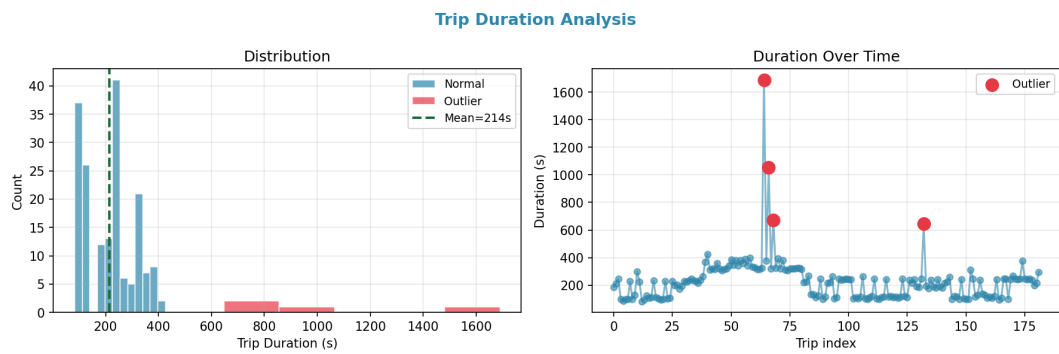


Figure 3.3: Overall duration distribution used for KPI extraction.

Chapter 4

Dispatch Policy Evaluation

Latest full-fleet result: 279 trips, 12-truck era.

4.1 Context: increased trucks and trips

Earlier analysis sat on a **small-fleet** history (many sessions with only **2–3 trucks**). Collection was expanded to `truck_count = 12`, the raw pool grew to **485** trips, then **279** full-fleet-era trips were kept for ranking.

Table 4.1: Analysis set used for policy ranking

Item	Value
Completed trips (decision events)	279
Fleet scale	9–12 trucks per session
Sessions	030256, 161650, 200004, 011337
Load / dump zones	10 / 14
Mean trip duration (logged)	~327 s
Mean trip fuel (logged)	~1.08 L
Coal moved (all policies)	257 t

Conclusion. Ranking reflects a **busy pit**. All six policies see the **same 279 dispatch moments** (event-driven offline replay). Only assignments and estimated cycle economics differ.

4.2 How policies were scored

Method. `dispatch_comparator.py` freezes each historical “truck free → choose mine” moment and replays alternate rules. No live re-sim of traffic per policy.

Table 4.2: Policies evaluated

Policy	Plain idea
GREEDY_NEAREST	Cheap on travel + queue + cycle
ETA_MIN	Lowest predicted full-cycle time
ETA_AWARE	ETA_MIN + learned bias (global/mine/truck)
FIFO	Mine waiting longest for assignment
LOAD_BALANCE	Lightest recent workload at mine
ROUND_ROBIN	Fixed rotation through mines

Composite score (higher = better): min-max normalize, then weight throughput 40%, total fuel 25%, avg duration 20%, mine-balance std 15%. Duration MAE is **not** in the composite.

4.3 Master score table

Table 4.3: Policy scores from `dispatch_policy_scores.csv` (279 trips each)

R	Policy	Comp.	T/hr	Dur	Fuel	CO ₂	Bal.	F/tr	Match
1	ETA_AWARE	60.0	6.34	198.2	261.6	701.1	11.03	0.938	64.9
2	FIFO	38.4	5.96	259.5	241.2	646.4	10.15	0.865	64.5
2	LOAD_BALANCE	38.4	5.96	259.5	241.2	646.4	10.15	0.865	64.5
4	ROUND_ROBIN	32.3	5.96	259.2	240.9	645.7	10.53	0.864	64.5
5	ETA_MIN	31.7	5.96	258.1	239.9	642.9	10.66	0.860	61.6
6	GREEDY_NEAREST	29.9	5.96	258.0	239.8	642.7	10.77	0.860	60.6

Shared across all policies: **total_trips = 279**, **total_coal = 257.0 t**, **coal/trip $\approx$ 921 kg**.

4.4 Best by metric and deltas vs FIFO

Table 4.4: Best policy by individual metric

Metric	Winner	Value
Composite	ETA_AWARE	60.0
Throughput	ETA_AWARE	6.34 trips/hr
Avg duration	ETA_AWARE	198.2 s
Total fuel	GREEDY_NEAREST	239.82 L
Mine balance	FIFO / LOAD_BALANCE	10.15
Duration MAE	FIFO / LOAD_BALANCE	120.9 s

Table 4.5: Relative deltas vs FIFO baseline

Policy	Δ trips/hr	Δ duration	Δ fuel
ETA_AWARE	+6.4%	−23.6%	+8.5%
LOAD_BALANCE	0%	0%	0%
ROUND_ROBIN	0%	−0.1%	−0.1%
ETA_MIN	0%	−0.5%	−0.5%
GREEDY_NEAREST	0%	−0.6%	−0.6%

Read: only **ETA_AWARE** moves the productivity needle; other rules cluster together.

4.5 Assignment patterns and disagreement

All policies still favor `load_zone_3` (~ 17.6 – 17.9%). `ETA_AWARE` differs more on dumps (`dump_zone_3` $\sim 22.6\%$ vs others $\sim \text{dump_zone_1/2}$). Gains come from cycle pairing / cost estimates, not abandoning the popular mine.

Two families: fairness/rotation (`FIFO` $\equiv$ `LOAD_BALANCE`) vs `ETA`/travel (`GREEDY` $\approx$ `ETA_MIN`; `ETA_AWARE` distinct).

Table 4.6: Pairwise mine-choice disagreement (%)

	ETA_AW	ETA_MIN	FIFO	GREEDY	LB	RR
ETA_AWARE	0	25.8	38.0	30.8	38.0	38.0
ETA_MIN	25.8	0	37.3	15.1	37.3	36.9
FIFO	38.0	37.3	0	36.6	0.0	6.5
GREEDY	30.8	15.1	36.6	0	36.6	35.8
LOAD_BALANCE	38.0	37.3	0.0	36.6	0	6.5
ROUND_ROBIN	38.0	36.9	6.5	35.8	6.5	0

4.6 Policy figures

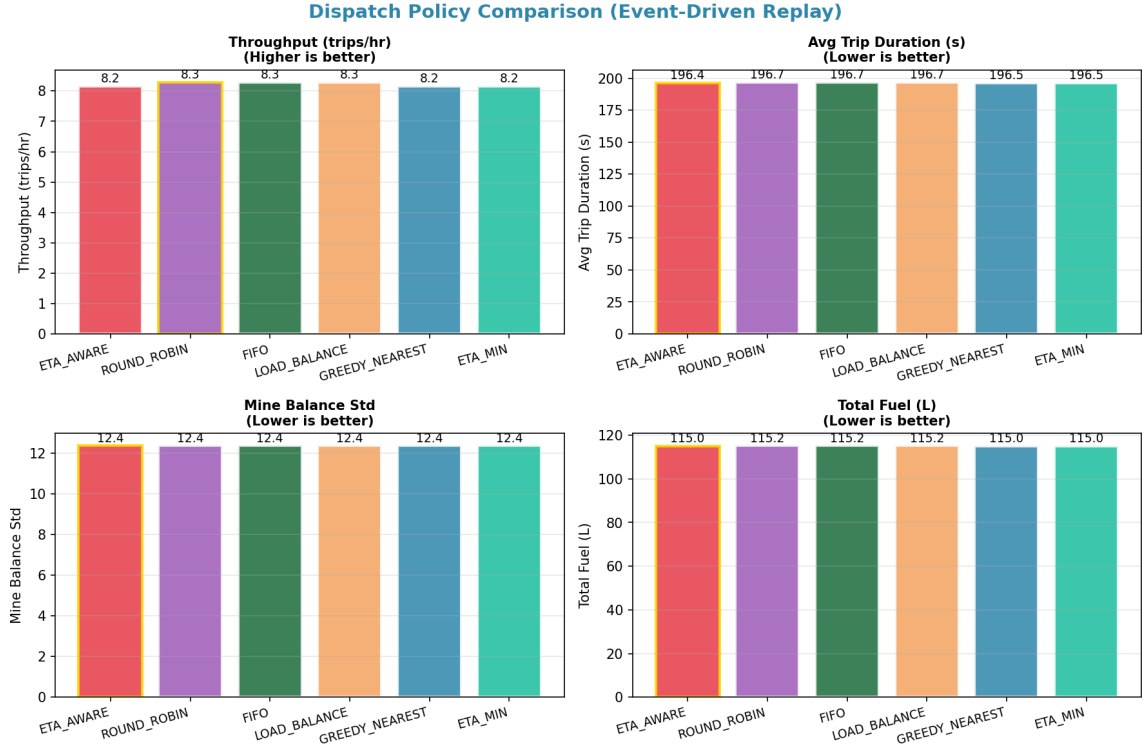


Figure 4.1: Composite KPI comparison across the six dispatch policies.

4.7 Chapter conclusions

1. Scaling to **279 full-fleet trips** is the right evidence base.
2. **ETA_AWARE** wins composite (60): best throughput and shortest cycles.
3. Cost: ~8–9% more fuel vs greedy/FIFO-family.
4. FIFO and LOAD_BALANCE are identical here (0% disagreement).
5. Other policies tie at 5.96 trips/hr; fuel/duration gaps are small.
6. Offline replay caveat: not closed-loop live A/B; MAE $\neq$ operational rank.

Bottom line: favor **ETA_AWARE** for hauls/hour and cycle speed; use **GREEDY/ETA_MIN** if fuel budget dominates.



Figure 4.2: Pairwise disagreement: how often two policies choose different mines.

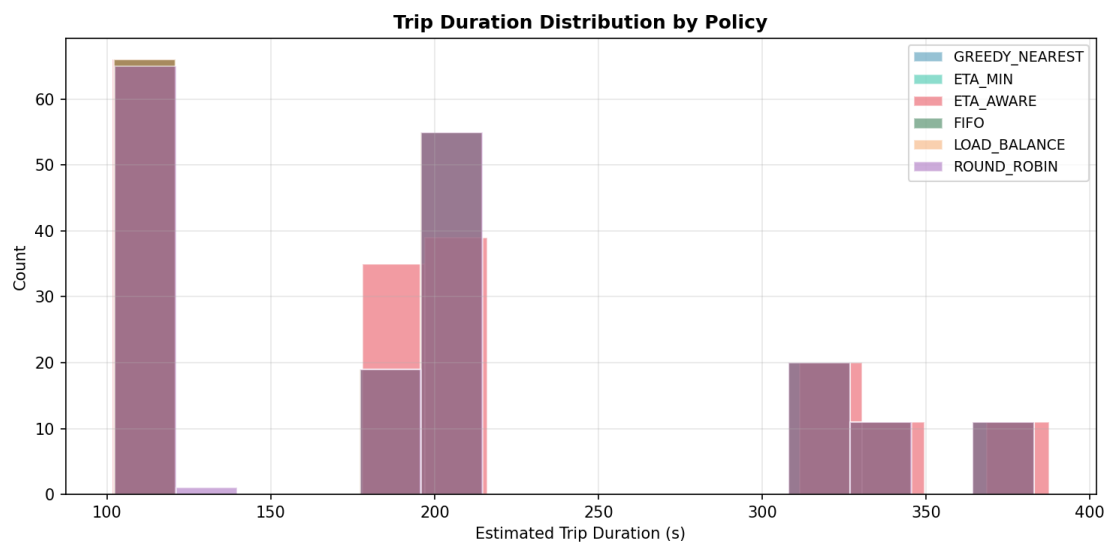


Figure 4.3: Estimated cycle-duration distributions by policy.

Chapter 5

Preview: EDA and ETA plots

The following figures already exist from earlier EDA/ETA work. Shown here so the report package stays complete.

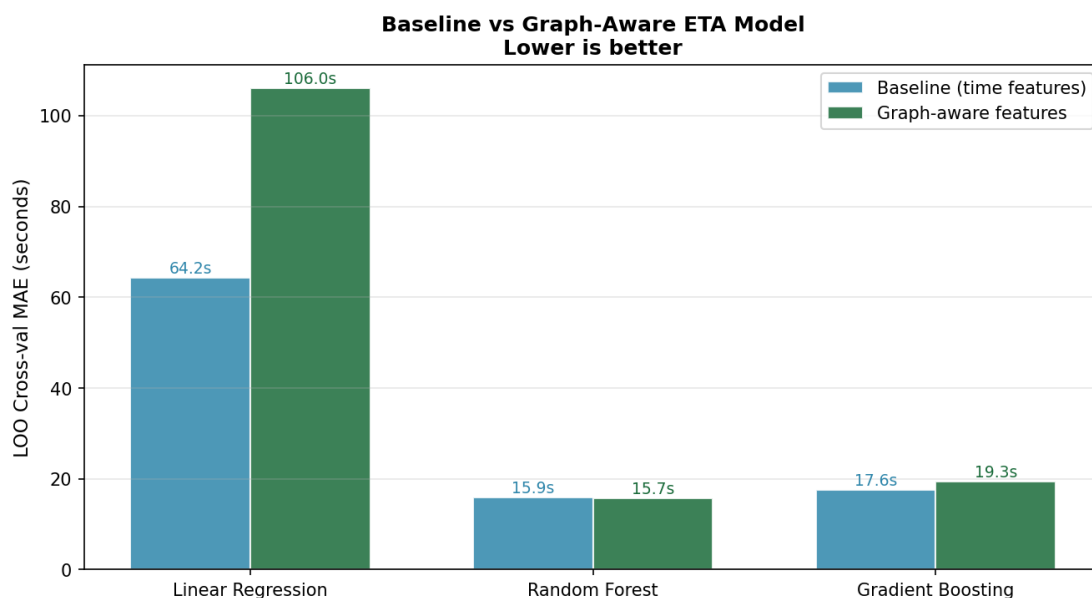


Figure 5.1: Baseline vs graph-aware ETA features.

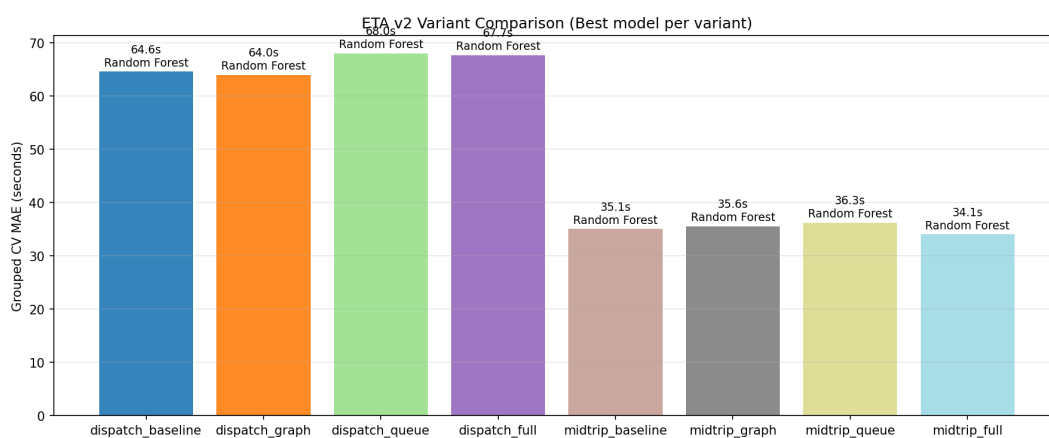


Figure 5.2: ETA v2 variant comparison (MAE).

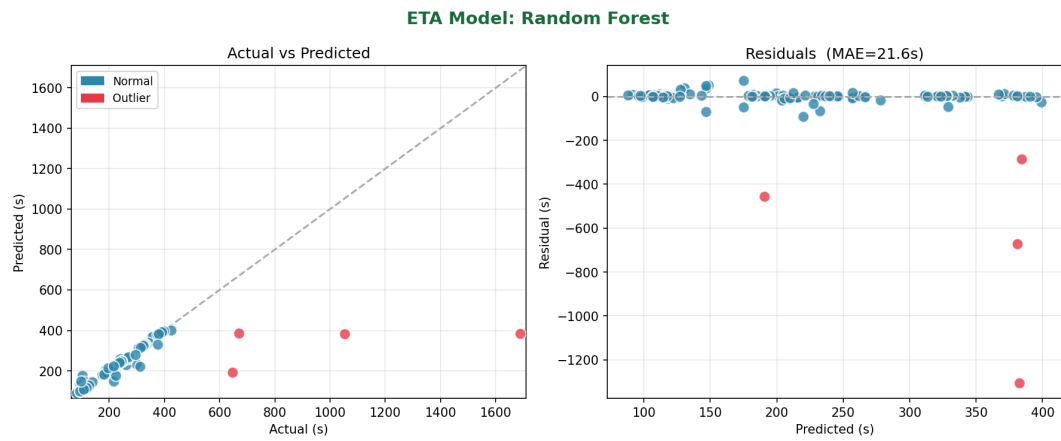


Figure 5.3: Random-forest ETA fit / residuals summary.

Chapter 6

Conclusions and Future Work

6.1 What we answered

This project built a full offline analytics path on simulation logs: describe operations as trips, predict cycle time with ETA models, and prescribe by scoring alternate dispatch rules on the same history.

On the locked **279**-trip full-fleet set (9–12 trucks), the core question has a concrete answer: logged traces are enough to measure, forecast, and **rank** policies without live A/B in the closed loop.

6.2 Main findings

1. **Data.** Prefer a homogeneous busy-fleet sample over a larger mixed pool: **279** full-fleet trips beat **485** mixed rows for policy and congestion claims.
2. **Predict.** Graph-aware and learned ETA variants give dispatch-time duration estimates that feed policy scoring (especially **ETA_MIN** / **ETA_AWARE**).
3. **Prescribe.** Under the stated composite weights, **ETA_AWARE** ranks first (composite 60): highest throughput (6.34 trips/hr) and shortest estimated cycles (~ 198 s), at roughly 8–9% higher fuel than the greedy/FIFO family.
4. **Trade-off.** If diesel budget dominates, prefer **GREEDY_NEAREST** or **ETA_MIN**. FIFO and **LOAD_BALANCE** are identical here and remain the fairness baseline.
5. **Caveat.** Offline event-driven replay freezes traffic history; it does not re-simulate multi-agent congestion under each policy. Duration MAE alone does not overturn operational KPI ranking.

6.3 Limitations

- Simulation-only evidence; not a real-mine deployment study.
- Continuous metrics (fuel, path length) are noisier than trip counts because soft path-following can leave the painted road.
- ETA sample size and extreme congestion coverage remain moderate.
- Policy deltas among non-**ETA_AWARE** rules are small and should be read as directional.

6.4 Future work

1. **Closed-loop A/B.** Run each policy live in the simulator so queues and interference update with the chosen assignments.

2. **Richer ETA.** Stronger queue features, mid-trip ETA for rescheduling, and wider route/-congestion coverage.
3. **Scale and robustness.** More full-fleet sessions, stress tests at higher truck counts, and sensitivity of ranking to composite weights.
4. **Operations bridge.** Package the offline comparator as an online decision aid (recommend mine + show predicted cycle/fuel) before any field pilot.

Closing line. The stack works end-to-end on simulation data: clean trips $\rightarrow$ ETA $\rightarrow$ ranked dispatch. Next step is closing the loop inside the sim, then proving the same ranking under live traffic feedback.